

Week 4 Lab / Activity Handout

EDA Design Flow Integration: Sequential FIR RTL and Simulation

1. Purpose of This Activity

This activity converts the Week 3 FIR architecture into the Week 4 baseline sequential RTL. The focus is correctness: clocked d0/d1/d2 sample history, active-low reset, valid-controlled acceptance, combinational FIR arithmetic, and testbench/waveform evidence. The activity is software-only and prepares students for both later implementation flows.

2. Learning Goals

- Relate the 3-tap FIR equation to the d0/d1/d2 clocked register chain.
- Run the self-checking fir3_seq simulation.
- Explain reset release, valid hold behavior, and post-edge sample history.
- Inspect PASS/FAIL output and waveform evidence.
- Explain one output sample from d0/d1/d2 and the FIR equation.
- Prepare for Week 5 parallelism and pipelining.

3. Course Scoring Reminder

Component	Weight	Week 4 connection
Xilinx FPGA design-flow evaluation project	45%	Builds the RTL/simulation evidence required before later synthesis and implementation.
Cell-based digital IC design-flow evaluation project	45%	Builds clean synthesizable RTL before standard-cell synthesis and timing analysis.
Attendance and participation	10%	Activity completion, discussion, checkpoint questions, and evidence submission.

4. Activity Schedule for Today

Time	Activity	Expected result
0-80 min	Teaching slides: sequential FIR, d0/d1/d2, reset, valid timing, waveform evidence	You can explain how the baseline FIR changes across accepted edges.
80-90 min	Break	Short pause.
90-155 min	Guided lab activity: Parts A-F	You run the source package and record evidence.
155-165 min	Break	Short pause.
165-180 min	Checkpoint, debugging notes, Week 5 preparation	You confirm correctness evidence and know what to review.

5. Software-Only Scope for Week 4

- Software simulation only.
- No FPGA board execution is required.
- No ASIC tapeout, fabrication, or silicon validation is involved.
- Evidence is source code, simulation log text, selected waveform signals, and short explanations.

6. Files Used in This Lab

File	Purpose	Student action
rtl/fir3_seq.v	Sequential d0/d1/d2 state + combinational FIR arithmetic	Read the RTL and identify state, reset, valid, and arithmetic nodes.
tb/tb_fir3_seq.v	Main self-checking testbench	Run simulation and inspect PASS/FAIL messages.
fir3_expected_outputs.txt	Reference accepted-sample outputs and exact testbench timing	Compare hand calculations and waveform states.
scripts/run_xsim_direct.sh	Vivado/XSim run helper	Run after sourcing Vivado settings.
scripts/run_icarus.sh	Icarus Verilog run helper	Optional alternative simulator flow.
tb/tb_fir3_seq_signed.v	Optional signed-width boundary sanity test	Instructor/reference use; not required for the main activity.

Each Verilog file includes `timescale 1ns/1ps` at the top. The main source uses the finalized course convention: $d0=x[n]$, $d1=x[n-1]$, $d2=x[n-2]$ after an accepted rising edge, with combinational y_out .

7. Part A: Review the 3-Tap FIR Equation

Use $h0=1$, $h1=2$, $h2=1$. Assume $d0=d1=d2=0$ before the first accepted sample.

$$y[n] = 1 \cdot x[n] + 2 \cdot x[n-1] + 1 \cdot x[n-2]$$

Accepted index	$x[n]$	$x[n-1]$	$x[n-2]$	Calculation	Expected $y[n]$
0	1	0	0	$1 \cdot 1 + 2 \cdot 0 + 1 \cdot 0$	1
1	2	1	0	$1 \cdot 2 + 2 \cdot 1 + 1 \cdot 0$	4
2	3	2	1	$1 \cdot 3 + 2 \cdot 2 + 1 \cdot 1$	8
3	4	3	2	$1 \cdot 4 + 2 \cdot 3 + 1 \cdot 2$	12
4	5	4	3	$1 \cdot 5 + 2 \cdot 4 + 1 \cdot 3$	16

8. Part B: Read the RTL Before Simulation

Question	Your answer
What are the input ports?	
What are the output ports?	
Which registers store accepted sample history?	
What does $rst_n=0$ do?	
What happens at a rising edge when $valid_in=0$?	
Which assignments perform $d0/d1/d2$ shifting?	
Which line makes y_out combinational?	
Why is $valid_out$ a reg but y_out a wire?	

9. Part C: Run the Simulation

Use the instructor-provided script if available. Otherwise, from the package root:

```
xvlog -nolog rtl/fir3_seq.v tb/tb_fir3_seq.v
xelab -nolog tb_fir3_seq -debug typical -s tb_fir3_seq_sim
xsim -nolog tb_fir3_seq_sim -runall
```

Expected final line: [RESULT] FIR3_SEQ TEST PASSED

10. Part D: Waveform Reading Questions

Question 1: At what simulation time is rst_n released? Does that release capture x_in?

Question 2: At which rising edges are the accepted samples 1, 2, 3, 4, and 5 captured?

Question 3: What happens at the 55 ns rising edge when x_in=99 and valid_in=0?

Question 4: Immediately before the 65 ns edge that accepts x_in=3, what are d0, d1, and d2? What are they immediately after the edge?

Question 5: Which d0/d1/d2 values and arithmetic prove that y_out=8 is correct?

Question 6: Which cycles should the testbench compare y_out, and why?

11. Part E: Debugging Checklist

Symptom	Possible cause	Check first
Output is X	Reset did not initialize state or signal connectivity is wrong	rst_n, d0, d1, d2, y_out.
Output sequence is shifted	Accepted-edge/valid timing mismatch	valid_in, valid_out, rising-edge times.
Values are duplicated	Wrong register update semantics	d0/d1/d2 before and after an accepted edge; confirm <=.
Invalid sample changed history	Hold behavior is wrong	At valid_in=0 edge, verify d0/d1/d2 do not change.
Arithmetic mismatch	Wrong sample/coefficient path	p0, p1, p2, s0, y_out.
Testbench fails on a held cycle	Output checked while valid_out=0	Comparison condition and valid_out.

12. Part F: Evidence Checklist

Evidence item	Required?	Your note
fir3_seq.v and tb_fir3_seq.v	Yes	
Simulation log containing PASS/FAIL and final RESULT line	Yes	
Waveform with clk, rst_n, valid_in, x_in, d0, d1, d2, valid_out, y_out	Yes	
One observation showing invalid x_in=99 does not shift sample history	Yes	
One hand explanation of a valid output, such as y_out=8	Yes	
p0/p1/p2/s0 waveform nodes	Optional but useful for debugging	

13. Exit Reflection

Why does the FIR design need d0/d1/d2 registers?

Why does reset make the first outputs predictable?

Why does reset release itself not count as sample capture?

Why must valid_out be included in the waveform?

Why is y_out combinational in this Week 4 baseline?

What should you review before Week 5 begins parallelism and pipelining?

14. Preparation for Week 5

Review d0/d1/d2 sample-history movement, nonblocking assignments, active-low reset, valid-controlled hold behavior, the 1/4/8/12/16 reference outputs, and waveform evidence. Week 5 will change the hardware arrangement to study throughput and timing while preserving the intended FIR mathematics.